



OPTIVIR CRM • ENTERPRISE OPERATING SYSTEM

COMPLETE ENTERPRISE UI ARCHITECTURE

Comprehensive Functional Map, Hierarchical Tree & System Module Breakdown

Suite v4.8

Next.js 16 (Turbopack)

TypeScript

14 Core Workspaces

Verified 0 Errors

Executive Overview & Purpose

This document codifies the full front-end and full-stack architecture of the OptiVir Ads CRM platform. Built specifically for high-velocity performance marketing and enterprise client growth management, the application unifies 14 major operating modules across 18 interactive views with complete INR (¹) commercial tracking, 70/30 split productivity panels, and live multi-tenant telemetry.

1. Master UI Tree Structure (14 Modules)

OPTIVIR ADS CRM – SYSTEM ROOT

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%% % 01. FOUNDATION

% % % % Design System (Tokens: Dark Navy #0A1628, Crimson #B91C1C, Slate #E2E8F0)

% % % % Application Shell (Responsive Two-Column Container, 256px Collapsible Sidebar)

% % % % Global Navigation (5 Main Sections, Route Indicators, Badge Counters)

% % % % Global Search & Command Palette (Ctrl+K / # K Universal Search across all 17 entities)

% % % % Create Menu (Instant Creation Drawer for Leads, Invoices, Proposals & Tasks)

% % % % Notifications Center (Dynamic Bell Badge, Live Stream Status & Severity Drawer)

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%% % 02. DASHBOARD

% % % % Management Dashboard (Executive Revenue: ₹48,250, 142 Active Clients, 1,482 Leads)

% % % % Sales Dashboard (7-Stage Deal Funnel Progression with Weighted INR ₹ Values)

% % % % Marketing Dashboard (Lead Acquisition Attribution Breakdown Donut Chart)

% % % % Finance Dashboard (Spline Revenue Projection: Revenue, ARR, Cash Flow Curves)

% % % % My Dashboard (Personal Priority Agenda, Active Tasks Stopwatch, Operational Friction Review)

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%% % 03. CRM

% % % % Leads (List, Lead Detail, BANT Qualification Flyout, Lead Score 0-100, 1-Click Conversion)

% % % % Contacts (Decision-Maker Directory, Role Badges, Direct Email/Phone Actions)

% % % % Companies (Account Tiers, Health Factors: Active/At-Risk/Prospect, Renewal Timelines)

% % % % Opportunities (Weighted Revenue Forecast Ledger, Win Velocity: 24.8%, Stage Progression)

% % % % Sales Pipeline (6-Column Kanban Board with ACV in ₹ Lakhs, Velocity Warnings)

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%% % 04. CLIENT MANAGEMENT

% % % % Clients Directory (Enterprise Gold, Silver SLA, Growth Retainer, Multi-Tenant Filter)

% % % % Client 360° Profile (94/100 Health Strip, Active SOWs, Retainer Burn Gauge, Renewal Hub)

% % % % Client Onboarding (Sales Handoff, Assets Collection, Strategy & SOW Kick-Off)

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%% % 05. PROJECT OPERATIONS

% % % % Projects Cockpit (List / Kanban / Timeline Views, Budget Burn Progress, Milestones Delivery)

% % % % Tasks & Detail (70/30 Split View #OPT-9492, Interactive Stopwatch, Subtasks, Dependencies)

% % % % Unified Calendar (Month Grid, Today Highlights, Mini Navigator, Agenda with 'Join Meet')

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%% % 06. SALES & DOCUMENTS

% % % % Proposals (Proposals Ledger, 7 KPI Summary Cards, Value Tracking)

% % % % 3-Column Proposal Builder (Document Outline, A4 Live Canvas Fig 4.1, Security Controls)

% % % % Quotations (PDF Preview, Commercial Terms, 1-Click Convert to Tax Invoice)

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%% % 07. FINANCE

% % % % Finance Ledger (Gross Revenue ₹48,250, Receivables ₹14,200, Net Margin 34.2%)

% % % % Invoices (Tax Invoices Ledger, Critical AR Dunning Banner, GSTIN Breakdown)

% % % % Payments (ICICI NEFT Batch Settlements PAY-2026-118, Record Payment Modal)

% % % % Expenses (Retainer Burn vs Operational Paid Media Budget Allocations)

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%% % 08. MARKETING

% % % % Campaigns Fleet (Multi-Channel Fleet List, Launch Campaign Modal, Pacing Flags)

% % % % Marketing Intelligence (5 Anomaly Alerts, 10 KPI Ceilings, Blended ROAS 3.83x, 4 Charts)

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%% % 09. REPORTING & BI

% % % % Reports Center (High-Density Dossiers Ledger, Category Filters, Batch Dispatch)

% % % % Live Report Builder (Compilation Pipeline #GEN-9902 at 78%, Live Stream Terminal)

% % % % Executive Dossier Preview (Print-Ready QBR Report Modal with Financial Projections)

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%% % 10. DOCUMENT MANAGEMENT

% % % % Document Vault (1.84 GB Encrypted Storage Telemetry, Category Filtering)

% % % % Preview & Audit Drawer (DOC-2026-089-MSA, SHA-256 Hash Verification, DocuSign Status)

% % % % Entity Binding (Direct Association of Signed Assets to Deals, Projects & Invoices)

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%% % 11. ACTIVITIES & AUDIT

% % % % Activity Center (Cross-Functional Audit Log, Chronological Milestone Triggers)

% % % % Log Actions (Quick Modals: Log Call, Log Meeting, Log Email, Add Note, Add Follow-up)

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%% % 12. NOTIFICATIONS CENTER

% % % % Live Stream Status (Node-US-04 Active Socket Sync, 3 KPI Summary Cards)

% % % % Severity Categorization (Critical Finance, Commercial Win, Task Escalation, System Sync)

% % % % Floating Right Drawer (7 Unread Updates with 1-Click Contextual Actions, Zero-Backlog State)

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%% % 13. SETTINGS & ADMINISTRATION

% % % % Organization & Brand Core (Tenant UUID opt-9021-acme-prod, Logo Asset Box, Legal Metadata)

% % % % Users & Roles (42 / 50 Active Seats, Granular RBAC Permissions Matrix)

% % % % Regionalization & FX (Asia/Kolkata Timezone, Base INR ₹ Pegging, Auto Regional Shift)

% % % % System Preferences (Console Density: Dense/Comfortable, Auditory Alerts, Telemetry Diff)

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%% % 14. SYSTEM / GLOBAL UI ARCHITECTURE

- %% % Modals & Drawers (Quick Create Drawers, Confirmations, Delete Modals, Unsaved Changes Guard)
- %% % Dynamic States (Empty Zero-Backlog, Loading Skeleton Shimmers, Error 404/500 Fallbacks)
- %% % Responsive Layouts (Collapsible 256px Sidebar, Mobile Flyout Drawer, Fluid Viewports)

2. Comprehensive Module Directory & Implementation Specifications

01 FOUNDATION & APPLICATION SHELL

Components: frontend/src/app/page.tsx, Sidebar.tsx, Header.tsx

- Design System: Dark Navy (#0A1628), Crimson (#B91C1C), Slate (#E2E6EC), high-contrast data tables.
- Application Shell: 256px collapsible sidebar with agency badge, active workspace indicators.
- Global Navigation: 5 grouped suites (Overview, CRM, Delivery, Revenue, Operations).
- Global Search (# K): Instant fuzzy search modal indexing all 17 CRM entity types.
- Create Menu & Quick Actions: Instant slide-over forms for rapid lead, opportunity, and invoice entry.

02 DASHBOARD & EXECUTIVE COMMAND

Components: frontend/src/components/dashboard/DashboardView.tsx

- Executive Metrics: Total Revenue (₹48,250, +12.5%), Active Clients (142, +8.2%), Pipeline Leads (1,482).
- View Switcher: 5 Sub-dashboards (Management, Sales Pipeline, Marketing ROAS, Finance Cashflow, My Dashboard).
- Operational Friction Matrix: Alert ribbon with Overdue Tasks (4), Unpaid Invoices (₹14,200), At-Risk Projects (1).
- Interactive Charts: Comparative revenue spline curves and lead ingestion donut chart.

03 CRM (LEADS, CONTACTS, COMPANIES, OPPS, PIPELINE)

Components: LeadsView.tsx, LeadDetailView.tsx, ContactsView.tsx, CompaniesView.tsx, OpportunitiesView.tsx, PipelineView.tsx

- Leads Workspace: Lead Fit Score gauges (0-100), slide-over BANT qualification drawer, 1-click deal conversion.
- Contacts & Decision-Makers: Verified stakeholder records with direct email/phone channels.
- Companies Directory: Account tiers, health factors (Active / At-Risk / Prospect), ARR indicators.
- Opportunities Ledger: Weighted INR closing projections with stage probability calculations.
- Sales Pipeline Kanban: 6 visual deal stages with ACV in ₹ Lakhs and deal velocity monitors.

04 CLIENT MANAGEMENT & 360° PROFILES

Components: ClientsListView.tsx, Client360View.tsx

- Clients Directory: Enterprise Gold, Silver SLA, and Growth Retainer service tier segmentation.
- Client 360° Profile: Acme Technologies dossier featuring 94/100 client health factor strip.
- Account Performance: Live retainers burn gauge, open deliverables, active contracts, and renewal center.
- Onboarding Pipeline: Multi-stage tracking from sales handoff to SOW kick-off.

05 PROJECT OPERATIONS & 70/30 SPLIT TASKS

Components: ProjectsView.tsx, TasksView.tsx, CalendarView.tsx

- Projects Cockpit: Multi-mode views (Table, Kanban, Timeline) with milestone status and budget tracking.
- 70/30 Tasks Split View (#OPT-9492): Real-time interactive stopwatch, subtasks checklist, dependencies.
- Unified Calendar: Monthly grid with today highlighted (Sep 10), agenda drawer, and Google Meet integration.

06 SALES & PROPOSAL BUILDER

Components: ProposalsView.tsx

- Proposals Ledger: Status lifecycle (Draft, Sent, Under Review, Approved, Signed).
- 3-Column Proposal Builder: Left outline & token vault, Center A4 live page canvas with Figure 4.1 diagram, Right security & signing controls.
- Quotations: Instant PDF preview generation and 1-click conversion to Tax Invoice.

07 FINANCE & BILLING LEDGER**Components:** FinanceView.tsx

- Finance KPIs: Gross billing, cash collected, and outstanding receivables in INR (₹).
- Invoices Engine: Detailed invoice ledger, net margin telemetry (34.2%), and AR dunning banner.
- Payments Settlements: ICICI NEFT batch sync ledger matching settlement references (PAY-2026-118).
- Expenses: Agency media spend vs client retainer burn tracking.

08 MARKETING INTELLIGENCE & ROAS**Components:** MarketingView.tsx

- Anomaly Detection: 5 automated algorithmic alerts (Critical, Efficiency, Pacing, Release, Optimal).
- 10 Growth KPIs: Blended ROAS (3.83x), Spend (₹64.80L), Revenue Attributed (₹2.48 Cr), CTR, CPA.
- Omnichannel Matrix: Cross-channel breakdown across Google, Meta, LinkedIn, DV360, and Microsoft Ads.

09 REPORTING & BI BUILDER**Components:** ReportsView.tsx, ExecutiveReportModal.tsx

- Reports Center: High-density dossiers ledger, scheduled reports, and batch export actions.
- Live Report Builder: Compilation pipeline #GEN-9902 at 78% completion with real-time log inspector.
- Executive Dossier: Print-ready quarterly business review (QBR) modal.

10 DOCUMENT MANAGEMENT VAULT**Components:** DocumentsView.tsx

- Encrypted Storage: 1.84 GB encrypted storage telemetry with category distribution progress bars.
- Audit & Preview Drawer: SHA-256 validation hash and DocuSign certificate inspector for DOC-2026-089-MSA.
- Entity Binding: Direct association of contracts to CRM deals, projects, and invoices.

11 ACTIVITIES & COMMUNICATIONS TIMELINE**Components:** CalendarView.tsx

- Cross-Functional Audit Log: Unified chronological stream of client engagements and system actions.
- Activity Logging: Fast modals to log phone calls, meetings, emails, internal notes, and follow-ups.

12 NOTIFICATIONS CENTER & LIVE STREAM**Components:** NotificationsView.tsx

- Real-time Stream: Active websocket telemetry indicator (Node-US-04) with unread priority badge.
- Severity Matrix: Color-coded categorization (Critical Finance, Commercial Win, Task Escalation, System Sync).
- Floating Right Drawer: 7 unread updates with 1-click contextual actions and zero-backlog toggle.

13 SETTINGS & SYSTEM ADMINISTRATION**Components:** SettingsView.tsx

- Organization Core: Tenant UUID opt-9021-acme-prod, master logo asset manager, legal company metadata.
- Users & Access: 42 / 50 active seats with granular role-based permissions matrix.
- Regionalization: Base currency INR (₹) with live FX pegging and Asia/Kolkata timezone handling.
- Preferences: Console density toggle (Dense / Comfortable), chime notifications, telemetry diffs.

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SYSTEM & GLOBAL UI INFRASTRUCTURE**Components:** Sidebar.tsx, Header.tsx, Layout Components

- Global Command Palette: Keyboard-first navigation (# K) to jump across any screen or action.
- Robust State Feedback: Contextual toast alerts, unsaved changes guards, and interactive error boundaries.
- Responsive Engine: Full mobile menu drawer, adaptive desktop column splits, and touch-friendly targets.

3. End-to-End Enterprise Operational State Flow

1. Inbound Acquisition**03. CRM — Leads**

Inbound leads ingestion via organic search, LinkedIn, and referral channels. Lead Fit Score gauges (0-100).

2. BANT Qualification**03. CRM — Opportunities**

Slide-over BANT qualification drawer checks Budget, Authority, Need, Timeline -> Promotes to Opportunity.

3. Proposal & SOW Drafting**06. Sales — Proposal Builder**

Creation in 3-Column Proposal Builder with Figure 4.1 system diagram and custom INR pricing tables.

4. Signature & Contract Close**10. Documents — Vault**

DocuSign integration with SHA-256 tamper verification hash -> Converts Opportunity to Active Client.

5. Client 360 Onboarding**04. Client Management**

Sales handoff, contract terms review, assets collection, and SLA tier assignment (Enterprise Gold 94/100).

6. Sprint Execution & 70/30 Tasks**05. Project Operations**

Project Cockpit milestones breakdown and 70/30 task detail split view with live stopwatch time tracking.

7. Milestone Tax Invoicing**07. Finance — Invoices**

Automatic invoice generation, GSTIN compliance, Net Margin tracking (34.2%), and AR Dunning.

8. Settlement Reconciliation**07. Finance — Payments**

Direct ICICI NEFT batch sync (PAY-2026-118) reconciles receivables into cash collected ledger.

9. Marketing Attribution & ROAS**08. Marketing Intelligence**

Cross-channel ad spend vs attributed client revenue analysis with blended ROAS benchmarking (3.83x).

10. Executive BI & QBR Reports**09. Reporting — Builder**

Compilation pipeline #GEN-9902 generates print-ready executive dossiers and client retention reviews.

Architecture Verification & Production Readiness

All 14 major modules and 18 interactive views have been verified in the OptiVir Ads application codebase.

- Turbopack Build: Completed with 0 TypeScript/Lint errors
- Server Environment: Next.js dev server on <http://localhost:3000> & Express backend on <http://localhost:5000>
- Native PDF Generation: Created via Pure JavaScript (pdfkit in Node.js)